

1 Solution — GIAM 1.5

Problem 3

1.1 Problem

Formulate a description of the **gcd** of two numbers in terms of their prime factorizations in the general case (when the factorizations may include powers of the primes involved).

1.2 The Description

Write each number using the **same list of primes** $p_1, p_2, \dots, p_k$ — every prime that divides *either* number — allowing exponent 0 for a prime that is absent:

$$a = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}, \quad b = p_1^{f_1} p_2^{f_2} \cdots p_k^{f_k}, \quad e_i, f_i \geq 0.$$

Then the greatest common divisor takes, for each prime, the **smaller** of the two exponents:

$$\gcd(a, b) = p_1^{\min(e_1, f_1)} p_2^{\min(e_2, f_2)} \cdots p_k^{\min(e_k, f_k)} = \prod_{i=1}^k p_i^{\min(e_i, f_i)}.$$

In words: the **gcd** is the **product over all primes of that prime raised to the smaller of its two exponents**.

1.3 Why the Minimum

A number $d = \prod_i p_i^{g_i}$ divides a exactly when its exponent on each prime does not exceed a 's — that is, $g_i \leq e_i$ for every i (this is what unique factorization buys us). For d to be a *common* divisor of a and b it must divide both, so

$$g_i \leq e_i \quad \text{and} \quad g_i \leq f_i, \quad \text{i.e.} \quad g_i \leq \min(e_i, f_i).$$

The **greatest** such d makes every exponent as large as allowed, namely $g_i = \min(e_i, f_i)$. That is precisely the boxed formula.

1.4 A Worked Example

Take $a = 2^3 \cdot 3 \cdot 5^2$ and $b = 2 \cdot 3^4 \cdot 7$. Listing both over the primes $\{2, 3, 5, 7\}$ (padding absent primes with exponent 0) and taking the smaller exponent in each column:

List both numbers over the SAME primes. For each prime, the gcd takes the smaller exponent.

prime p	exponent in a $a = 2^3 \cdot 3 \cdot 5^2$	exponent in b $b = 2 \cdot 3^4 \cdot 7$	min $\rightarrow$ in gcd
2	3	1	$\min(3,1) = 1$
3	1	4	$\min(1,4) = 1$
5	2	0	$\min(2,0) = 0$
7	0	1	$\min(0,1) = 0$

A prime missing from a number is included with exponent 0 (e.g. 5 is absent from b , 7 is absent from a).

$$\gcd(a, b) = \prod p^{\min(e_p, f_p)}$$

$$\text{Here: } \gcd = 2^1 \cdot 3^1 \cdot 5^0 \cdot 7^0 = 2 \cdot 3 = \mathbf{6}$$

lcm uses max instead of min: $\text{lcm}(a, b) = \prod p^{\max(e_p, f_p)}$. And $\min + \max = e_p + f_p$, so $\gcd \cdot \text{lcm} = a \cdot b$.

Figure 1.1: Both numbers are written over the same primes 2,3,5,7, padding an absent prime with exponent 0. For each prime the gcd takes the minimum of the two exponents: $\min(3,1)=1$ for 2, $\min(1,4)=1$ for 3, $\min(2,0)=0$ for 5, $\min(0,1)=0$ for 7. So $\gcd = 2 \cdot 3 = 6$. The lcm uses max instead, and $\min + \max = \text{sum}$ gives $\gcd \cdot \text{lcm} = ab$.

$$\gcd(a, b) = 2^{\min(3,1)} 3^{\min(1,4)} 5^{\min(2,0)} 7^{\min(0,1)} = 2^1 3^1 5^0 7^0 = 6.$$

1.5 Remark — Padding With Zero Exponents Is Essential

The phrase “the same list of primes” does the real work. A prime that appears in only one number is included in *both* factorizations with exponent 0 in the one it is missing from. Since $\min(e, 0) = 0$, such a prime contributes $p^0 = 1$ to the gcd — it correctly drops out. Without this padding convention the “minimum over each prime” rule would not be well defined, because the two products would range over different primes. This is exactly why, in Problem 2, the coprime pair $110 = 2 \cdot 5 \cdot 11$ and $273 = 3 \cdot 7 \cdot 13$ had $\gcd = 1$: *every* prime is missing from one of them, so every exponent-minimum is 0.

1.6 Remark — The Companion lcm Formula and the Product Identity

The same setup gives the least common multiple by swapping **min** for **max**:

$$\text{lcm}(a, b) = \prod_{i=1}^k p_i^{\max(e_i, f_i)}.$$

Because, for any two numbers, $\min(e, f) + \max(e, f) = e + f$, multiplying the two formulas prime-by-prime gives

$$\text{gcd}(a, b) \cdot \text{lcm}(a, b) = \prod_i p_i^{\min(e_i, f_i) + \max(e_i, f_i)} = \prod_i p_i^{e_i + f_i} = ab.$$

So the single identity $\text{gcd} \cdot \text{lcm} = ab$ used numerically in Problem 2 is just this exponent fact in disguise.

1.7 Remark — It Rests on Unique Factorization

This whole description is only meaningful because each number has *exactly one* prime factorization (up to order) — the Fundamental Theorem of Arithmetic. The exponents e_i, f_i are then unambiguous, the “divides $\iff$ exponent-wise $\leq$ ” criterion holds, and the **gcd** is pinned down uniquely. The formula also extends verbatim to more than two numbers: $\text{gcd}(a, b, c, \dots) = \prod_i p_i^{\min(e_i, f_i, g_i, \dots)}$, taking the minimum exponent across *all* the numbers.

1.8 Remark — Conceptual Clarity vs. Practical Computation

This min-of-exponents description is the clearest *conceptual* picture of what a **gcd** is, and it makes facts like $\text{gcd} \cdot \text{lcm} = ab$ transparent. It is not, however, how one usually *computes* a **gcd**: it presupposes the prime factorizations, and factoring large numbers is hard. Euclid’s algorithm (Problem 2) finds the same **gcd** without ever factoring. The two viewpoints are complementary — the prime-factorization formula tells us *what* the **gcd** is, while Euclid tells us *how* to get it quickly.